LAB 5 / SOFTWARE ENGINEERING

Static Code Analysis

Flight Reservation System

Pylint | Bandit | Flake8

Before and after analysis, findings, fixes, and reflection

Prepared from the lab handout and saved analyzer reports

Introduction

Static code analysis examines source code without running the application. It helps identify likely defects, unsafe coding patterns, and style inconsistencies early, while changes are still inexpensive to make. It complements runtime testing; it does not replace tests or review.

Pylint

A Python code-quality analyzer. It reports likely programming errors, weak practices, naming and documentation problems, and gives an overall score.

Bandit

A security-focused analyzer for Python. It scans for known risky constructs such as eval(), weak cryptographic hashes, and shell execution.

Flake8

A style and linting tool combining checks for PEP 8 formatting, pycodestyle, pyflakes, and McCabe complexity.

Scope: the original program was analyzed first; reported issues were then fixed and all three tools were rerun. Full saved outputs are reproduced in the appendices.

Before: Pylint findings

```
(base) PS C:\Users\Bhaga\Desktop\sem5\L5_50_flight_reservation_system-main> python -m pylint flight_reservation_system.py
************* Module flight_reservation_system
flight_reservation_system.py:23:0: C0301: Line too long (132/100) (line-too-long)
flight_reservation_system.py:87:0: C0304: Final newline missing (missing-final-newline)
flight_reservation_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
flight_reservation_system.py:12:0: C0103: Constant name "total_revenue" doesn't conform to UPPER_CASE naming style (invalid-name)
flight_reservation_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:15:0: C0103: Function name "bookFlightSeat" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
flight_reservation_system.py:16:4: W0603: Using the global statement (global-statement)
flight_reservation_system.py:23:24: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
flight_reservation_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:27:0: C0103: Function name "cancelBooking" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:31:4: W0702: No exception type(s) specified (bare-except)
flight_reservation_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:35:0: C0103: Function name "checkSeatAvailability" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:37:7: C0121: Comparison 'flights.get(flight_no) == None' should be 'flights.get(flight_no) is None' (singleton-
comparison)
flight_reservation_system.py:39:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
flight_reservation_system.py:39:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-ret
urn)
flight_reservation_system.py:36:4: W0612: Unused variable 'airport_fee' (unused-variable)
flight_reservation_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:45:0: C0103: Function name "calculateBaggageFee" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:47:16: W0123: Use of eval (eval-used)
flight_reservation_system.py:46:4: W0612: Unused variable 'base_fare' (unused-variable)
flight_reservation_system.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:51:0: C0103: Function name "generatePassengerHash" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:55:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:55:0: C0103: Function name "generateBoardingPin" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:59:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:59:0: C0103: Function name "exportFlightManifest" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:60:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
flight_reservation_system.py:60:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
flight_reservation_system.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:66:0: C0103: Function name "getFlightSummary" doesn't conform to snake_case naming style (invalid-name)
flight_reservation_system.py:66:21: W0621: Redefining name 'flights' from outer scope (line 6) (redefined-outer-name)
flight_reservation_system.py:68:8: W0612: Unused variable 'f' (unused-variable)
flight_reservation_system.py:74:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:4:0: W0611: Unused import math (unused-import)

-------------------------------------------------------------------
Your code has been rated at 4.00/10 (previous run: 4.00/10, +0.00)
```

After: Pylint results

```
(base) PS C:\Users\Bhaga\Desktop\sem5\SE\lab5\L5_50_flight_reservation_system> python -m pylint flig
ht_reservation_system.py

--------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

Before: Bandit findings

```
(base) PS C:\Users\Bhaga\Desktop\sem5\L5_50_flight_reservation_system-main> python -m bandit flight_reservation_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.7
Run started:2026-10-30 09:58:06.425437+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\flight_reservation_system.py:31:4
30              flights[flight_no]["is_full"] = False
31          except:
32              pass
33

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\flight_reservation_system.py:47:16
46          base_fare = flights[flight_no]["fare"]
47          final_fee = eval(surcharge_expr)
48          return final_fee

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\flight_reservation_system.py:52:11
51      def generatePassengerHash(passenger_id):
52          return hashlib.md5(passenger_id.encode()).hexdigest()
53

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\flight_reservation_system.py:56:11
55      def generateBoardingPin():
56          return random.randint(1000, 9999)
57

--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\flight_reservation_system.py:63:4
62          f.close()
63          os.system("echo Boarding manifest printed: " + filename)
64

--------------------------------------------------
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'AIRLINE_DISPATCH_SECRET_TOKEN_2026'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\flight_reservation_system.py:75:25
74      def main():
75          airline_api_secret = "AIRLINE_DISPATCH_SECRET_TOKEN_2026"
76          bookFlightSeat("AI101", "Kritika", 20)

--------------------------------------------------

Code scanned:
        Total lines of code: 65
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 3
                Medium: 1
                High: 2
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 1
                High: 5
Files skipped (0):
```

After: Bandit results

```
(base) PS C:\Users\Bhaga\Desktop\sem5\SE\lab5\L5_50_flight_reservation_system> python -m bandit -r f
light_reservation_system.py
[main]  INFO    profile include tests: None
[main]  INFO    profile exclude tests: None
[main]  INFO    cli include tests: None
[main]  INFO    cli exclude tests: None
[main]  INFO    running on Python 3.14.7
Run started:2026-10-30 09:57:19.874826+00:00

Test results:
        No issues identified.

Code scanned:
        Total lines of code: 132
        Total lines skipped (#nosec): 0

Run metrics:
        Total issues (by severity):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
        Total issues (by confidence):
                Undefined: 0
                Low: 0
                Medium: 0
                High: 0
Files skipped (0):
```

Before: Flake8 findings

```
(base) PS C:\Users\Bhaga\Desktop\sem5\L5_50_flight_reservation_system-main> python -m flake8 flight_reservation_system.py
flight_reservation_system.py:4:1: F401 'math' imported but unused
flight_reservation_system.py:7:80: E501 line too long (84 > 79 characters)
flight_reservation_system.py:8:80: E501 line too long (85 > 79 characters)
flight_reservation_system.py:9:80: E501 line too long (83 > 79 characters)
flight_reservation_system.py:23:80: E501 line too long (132 > 79 characters)
flight_reservation_system.py:31:5: E722 do not use bare 'except'
flight_reservation_system.py:36:5: F841 local variable 'airport_fee' is assigned to but never used
flight_reservation_system.py:37:31: E711 comparison to None should be 'if cond is None:'
flight_reservation_system.py:46:5: F841 local variable 'base_fare' is assigned to but never used
flight_reservation_system.py:87:7: W292 no newline at end of file
```

After: Flake8 results

```
(base) PS C:\Users\Bhaga\Desktop\sem5\SE\lab5\L5_50_flight_reservation_system> python -m flake8 flig
ht_reservation_system.py
(base) PS C:\Users\Bhaga\Desktop\sem5\SE\lab5\L5_50_flight_reservation_system>
```

Issue identification and fixes

Issue	Type	Line(s)	Description	Fix applied
Unsafe dynamic evaluation	Security	47	eval() could execute arbitrary input in the baggage fee calculation.	Replaced eval with a restricted AST evaluator allowing numeric literals, base_fare, and arithmetic operators only.
Weak MD5 hash	Security	52	MD5 is unsuitable for security-sensitive hashing because of known collision weaknesses.	Switched to hashlib.sha256().
Shell command execution	Security	63	os.system() executes through a shell and included a filename in the command string.	Removed shell invocation; write the manifest with a context manager and print status directly.

Line numbers refer to the original script analyzed by the tools.

Issue identification and fixes (continued)

Issue	Type	Line(s)	Description	Fix applied
Mutable default list	Bug	15	baggage_tags=[] is shared across calls, leaking data between bookings.	Changed the default to None and create a fresh list when needed.
Bare except / silent pass	Reliability	31-32	A catch-all handler hid errors and silently ignored invalid flight numbers.	Validate the flight number explicitly and return False for unknown flights.
Predictable PIN generator	Security	56	random.randint() is not cryptographically suitable for a boarding PIN.	Generate the PIN with secrets.randbelow().

Line numbers refer to the original script analyzed by the tools.

Reflection

1. What static analysis did you learn from this lab?

Static analysis can catch security and maintainability issues before runtime. Pylint highlighted quality and structure problems, Flake8 enforced formatting, and Bandit identified risky patterns including eval(), weak hashing, and shell execution.

2. Which issue was most important to fix, and why?

Unsafe eval() in the baggage surcharge calculation was the most important. If attacker-controlled input reached it, it could execute arbitrary code. Replacing it with a restricted arithmetic evaluator removed that execution risk.

3. How did the fixes improve code quality?

The fixes removed unsafe evaluation and shell use, replaced MD5 and the predictable PIN generator, avoided a shared mutable default, and made invalid flight handling explicit. The code also follows consistent naming and formatting.

4. Would you use static analysis in future projects?

Yes. Running these checks regularly helps find defects early, reduces debugging effort, improves security, and keeps a codebase consistent as it grows.

Appendix: Pylint before

Source file: pylint_report_before.txt

```
************* Module flight_reservation_system
flight_reservation_system.py:23:0: C0301: Line too long (132/100) (line-too-long)
flight_reservation_system.py:87:0: C0304: Final newline missing (missing-final-newline)
flight_reservation_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
flight_reservation_system.py:12:0: C0103: Constant name "total_revenue" doesn't conform to UPPER_CASE naming style
(invalid-name)
flight_reservation_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:15:0: C0103: Function name "bookFlightSeat" doesn't conform to snake_case naming style
(invalid-name)
flight_reservation_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
flight_reservation_system.py:16:4: W0603: Using the global statement (global-statement)
flight_reservation_system.py:23:24: C0209: Formatting a regular string which could be an f-string
(consider-using-f-string)
flight_reservation_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:27:0: C0103: Function name "cancelBooking" doesn't conform to snake_case naming style
(invalid-name)
flight_reservation_system.py:31:4: W0702: No exception type(s) specified (bare-except)
flight_reservation_system.py:35:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:35:0: C0103: Function name "checkSeatAvailability" doesn't conform to snake_case naming
style (invalid-name)
flight_reservation_system.py:37:7: C0121: Comparison 'flights.get(flight_no) == None' should be 'flights.get(flight_no)
is None' (singleton-comparison)
flight_reservation_system.py:39:4: R1703: The if statement can be replaced with 'return bool(test)'
(simplifiable-if-statement)
flight_reservation_system.py:39:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code
inside it (no-else-return)
flight_reservation_system.py:36:4: W0612: Unused variable 'airport_fee' (unused-variable)
flight_reservation_system.py:45:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:45:0: C0103: Function name "calculateBaggageFee" doesn't conform to snake_case naming style
(invalid-name)
flight_reservation_system.py:47:16: W0123: Use of eval (eval-used)
flight_reservation_system.py:46:4: W0612: Unused variable 'base_fare' (unused-variable)
flight_reservation_system.py:51:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:51:0: C0103: Function name "generatePassengerHash" doesn't conform to snake_case naming
style (invalid-name)
flight_reservation_system.py:55:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:55:0: C0103: Function name "generateBoardingPin" doesn't conform to snake_case naming style
(invalid-name)
flight_reservation_system.py:59:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:59:0: C0103: Function name "exportFlightManifest" doesn't conform to snake_case naming style
(invalid-name)
flight_reservation_system.py:60:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
flight_reservation_system.py:60:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
flight_reservation_system.py:66:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:66:0: C0103: Function name "getFlightSummary" doesn't conform to snake_case naming style
(invalid-name)
flight_reservation_system.py:66:21: W0621: Redefining name 'flights' from outer scope (line 6) (redefined-outer-name)
flight_reservation_system.py:68:8: W0612: Unused variable 'f' (unused-variable)
flight_reservation_system.py:74:0: C0116: Missing function or method docstring (missing-function-docstring)
flight_reservation_system.py:4:0: W0611: Unused import math (unused-import)

-----------------------------------
Your code has been rated at 4.00/10
```

Appendix: Bandit before (1)

Source file: bandit_report_before.txt

```
[main] INFO profile include tests: None
[main] INFO profile exclude tests: None
[main] INFO cli include tests: None
[main] INFO cli exclude tests: None
[main] INFO running on Python 3.12.10
Run started:2026-10-28 06:44:20.923356+00:00

Test results:
>> Issue: [B110:try_except_pass] Try, Except, Pass detected.
   Severity: Low   Confidence: High
   CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
   Location: .\flight_reservation_system.py:31:4
30         flights[flight_no]["is_full"] = False
31     except:
32         pass
33

--------------------------------------------------
>> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
   Severity: Medium   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
   Location: .\flight_reservation_system.py:47:16
46     base_fare = flights[flight_no]["fare"]
47     final_fee = eval(surcharge_expr)
48     return final_fee

--------------------------------------------------
>> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
   Severity: High   Confidence: High
   CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
   Location: .\flight_reservation_system.py:52:11
51 def generatePassengerHash(passenger_id):
52     return hashlib.md5(passenger_id.encode()).hexdigest()
53

--------------------------------------------------
>> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
   Severity: Low   Confidence: High
   CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
   Location: .\flight_reservation_system.py:56:11
55 def generateBoardingPin():
56     return random.randint(1000, 9999)
57
```

Appendix: Bandit before (2)

Source file: bandit_report_before.txt

```
--------------------------------------------------
>> Issue: [B605:start_process_with_a_shell] Starting a process with a shell, possible injection detected, security issue.
   Severity: High   Confidence: High
   CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b605_start_process_with_a_shell.html
   Location: .\flight_reservation_system.py:63:4
62     f.close()
63     os.system("echo Boarding manifest printed: " + filename)
64

--------------------------------------------------
>> Issue: [B105:hardcoded_password_string] Possible hardcoded password: 'AIRLINE_DISPATCH_SECRET_TOKEN_2026'
   Severity: Low   Confidence: Medium
   CWE: CWE-259 (https://cwe.mitre.org/data/definitions/259.html)
   More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b105_hardcoded_password_string.html
   Location: .\flight_reservation_system.py:75:25
74 def main():
75     airline_api_secret = "AIRLINE_DISPATCH_SECRET_TOKEN_2026"
76     bookFlightSeat("AI101", "Kritika", 20)

--------------------------------------------------

Code scanned:
 Total lines of code: 65
 Total lines skipped (#nosec): 0
 Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
 Total issues (by severity):
  Undefined: 0
  Low: 3
  Medium: 1
  High: 2
 Total issues (by confidence):
  Undefined: 0
  Low: 0
  Medium: 1
  High: 5
Files skipped (0):
```

Appendix: Flake8 before

Source file: flake8_report_before.txt

```
flight_reservation_system.py:4:1: F401 'math' imported but unused
flight_reservation_system.py:7:80: E501 line too long (84 > 79 characters)
flight_reservation_system.py:8:80: E501 line too long (85 > 79 characters)
flight_reservation_system.py:9:80: E501 line too long (83 > 79 characters)
flight_reservation_system.py:23:80: E501 line too long (132 > 79 characters)
flight_reservation_system.py:31:5: E722 do not use bare 'except'
flight_reservation_system.py:36:5: F841 local variable 'airport_fee' is assigned to but never used
flight_reservation_system.py:37:31: E711 comparison to None should be 'if cond is None:'
flight_reservation_system.py:46:5: F841 local variable 'base_fare' is assigned to but never used
flight_reservation_system.py:87:7: W292 no newline at end of file
```

Appendix: Pylint after

Source file: pylint_report_after.txt

```
-------------------------------------------------------------------
Your code has been rated at 10.00/10 (previous run: 10.00/10, +0.00)
```

Appendix: Bandit after

Source file: bandit_report_after.txt

```
[main] INFO profile include tests: None
[main] INFO profile exclude tests: None
[main] INFO cli include tests: None
[main] INFO cli exclude tests: None
[main] INFO running on Python 3.12.10
Run started:2026-10-28 06:47:16.610069+00:00

Test results:
 No issues identified.

Code scanned:
 Total lines of code: 132
 Total lines skipped (#nosec): 0
 Total potential issues skipped due to specifically being disabled (e.g., #nosec BXXX): 0

Run metrics:
 Total issues (by severity):
  Undefined: 0
  Low: 0
  Medium: 0
  High: 0
 Total issues (by confidence):
  Undefined: 0
  Low: 0
  Medium: 0
  High: 0
Files skipped (0):
```

Appendix: Flake8 after

Source file: flake8_report_after.txt

```
(The saved report file is empty. Flake8 emits no text when no violations are found.)
```